

DIGITAL PROJECT MANAGEMENT USING AI

NTW 22 JULY 2026

Welcome to National Training Week: 14 Pathways to Excellence

Alpha Pinnacle Consulting is proud to participate in NTW 2026 for the first time, delivering highly specialized training across Engineering, Construction, Project Management, and Advanced Technology.



Engineering & Construction

How Can I Become a Professional Engineer (Ir.) in Malaysia? A Step-by-Step Roadmap

Construction Planning: Kurangkan Kelewatan & Tingkatkan Keuntungan Kontraktor

Turning the Construction Design Management (CDM) 2024 Compliance Into a Strategic Business Advantage

ISO 9001 Awareness in the Construction Industry



Project Management

Beyond Time, Cost & Quality: PMP® Principles for Engineering & Construction Project Success

Pensijilan CIDB SKKP (CCPM, CCM & CSS): Meningkatkan Prestasi Perniagaan & Pengurusan Projek

Bagaimana Pensijilan CIDB SKKP FMM dan FME Boleh Membantu Kontraktor Pengurusan Fasilitas: Bhg. 1



AI & Data Analytics

Artificial Intelligence for Project Managers: Are You Ready?

Digital Project Management Using AI

AI For Business Productivity

Visual Analytics with Power BI & AI



Business Excellence

Zero Defect Mindset ke Arah Kualiti, Keuntungan & Daya Saing Kontraktor Bhg. 1

LENSA: Smarter Thinking for Better Problem Solving & Decision Making

Customer Service Excellence in Hospitality

AI in One Word

When you hear “AI and project management” together, what is the first word that comes to mind?

Efficiency

Uncertainty

Automation

Opportunity

Resistance

Speed

Hold one thought: a repetitive task in your current project. We return to it in Module 3.

MODULE

1



Foundations of AI in Project Management

Evolution of digital PM • AI/ML basics • AI in the PM lifecycle

Four Eras of Digital Project Management

1

Manual & Paper-Based

Static Gantt charts, manual reports, tacit knowledge

Whiteboards, spreadsheets

2

Desktop PM Software

Critical path, resource levelling on one machine

MS Project, Primavera P6

3

Cloud & Collaborative

Real-time collaboration, cross-team visibility

Jira, Asana, Primavera Cloud

4

AI-Augmented PM

Predictive insights, automated flags, NL reporting

AI copilots, predictive platforms

Most organisations today operate across a blend of these eras — the shift into Era 4 is the focus of this programme.

AI & ML in Plain Language



Artificial Intelligence

Systems performing tasks that normally require human judgement — patterns, predictions, recommendations.



Machine Learning

Learns patterns from historical data instead of following rules you program explicitly.



Predictive Analytics

Forecasts outcomes — e.g. the likelihood a task finishes late — from historical and live data.



Natural Language Processing

Reads, summarises, and generates human language — minutes, chat queries, draft reports.



Generative AI

Produces new content — summaries, draft risk registers, even draft schedules — from a prompt.

Key distinction: Automation follows fixed rules you define. AI/ML learns patterns and improves as more data arrives.

AI Across the PM Lifecycle



Trainer insert point: add a current, verified adoption statistic here (WEF / McKinsey / Gartner / Deloitte) — confirm before each delivery.

Levels of AI Usage in Project Management

↑ MORE AUTONOMY

LEVEL 4 Autonomous AI PMO

Multiple agents orchestrated end-to-end: monitoring, re-planning, and reporting across the portfolio

LEVEL 3 Agentic AI

AI plans and executes multi-step workflows toward a goal — e.g. detects slippage, drafts recovery options, updates the schedule

LEVEL 2 AI Agent

AI performs a defined task with tools on your behalf — e.g. transcribes the meeting, files minutes, creates action items

LEVEL 1 Prompt Engineering

Human asks, AI answers — one prompt, one output: summaries, drafts, risk brainstorm

Climb one level at a time: master prompting before deploying agents — each level up trades human control for autonomy, so oversight must scale with it.

Building Effective Project Context

Building the right context starts it all

WHO

E.g. BigEnergy an Energy Producing Company operating across global project sites with distributed project teams

WHY

E.g. Improve real-time remote collaboration and accelerate executive decision-making at the operations level

WHAT

E.g. Design and install 20 high tech LED panels to build a fully integrated command centre at the Kuala Lumpur HQ in 22nd floor of TRX building for live project monitoring

WHEN

E.g. Full installation and commissioning completed before December 2026 — aligned with the Grand Opening milestone

 Pro tip: Use 'Project' or Use the same context for every new AI session to maintain consistency across all your project documents.

MODULE

2

AI Tools & Applications

Predictive scheduling • Risk management • Resource allocation • Dashboards



Use Case – Starting the Project (Project Initiation)

Using ChatGPT

01

Paste your 4W project context

02

Enter your prompt

03

Review and refine the output

04

Export to Word document

"Create a detailed Project Charter including objectives, scope, stakeholders, and key risks. Format it as a structured document."

Using Google Docs

01

Click the Gemini Magic Wand

02

Enter your 4W project context prompt

03

Review the auto-formatted output

04

Apply your organization's template

"Generate a Project Charter based on this project context: [paste 4W]"



Tip: Always review AI output critically. Add your local context, correct assumptions, and ensure stakeholder names are accurate before sharing.

Use Case – Project Scoping (Project Planning)

A well-structured WBS is the backbone of every successful project. AI makes it easy to decompose complex scopes into manageable, organized tasks — in minutes.

How to Generate Your WBS

01

Prompt AI with your project context

"Break down this project into a full Work Breakdown Structure (WBS) with phases and tasks."

02

Refine the hierarchy

Level 1: Project → Level 2: Phases → Level 3: Tasks

03

Export and visualize

Use the WBS result for a structured table or Miro for a visual board

Tools that can be used

ChatGPT

Draft the WBS structure

Gemini

Use Google Sheets and Magic Wand

Miro or Gemini

To create a Visual hierarchy board

Use Case – Project Schedule (Project Planning)

Turn your WBS tasks into a visual timeline. AI accelerates schedule creation by estimating durations, sequencing tasks, and identifying dependencies — even flagging the critical path.



ChatGPT

Prompt AI to generate task durations and sequencing. Ask: *"Estimate realistic durations for each WBS task and identify dependencies."*



Google Sheets and Gemini

Use Gemini to convert your task list into a formatted Gantt chart with start/end dates, milestones, and color-coded phases.



Miro / Gemini

Build a collaborative visual timeline your whole team can interact with — great for kick-off meetings and sprint planning sessions.



Exercise: Create your project timeline and highlight the critical path — the sequence of tasks that directly determines your project end date.

All Sorts of Prompts

PROMPT for Risk Register

PROMPT 7 — Risk Register (12–15 risks)
Analyse this project for risks: [PROJECT DESCRIPTION + WBS SUMMARY]
Create a risk register covering: Technical | Schedule | Resource | Stakeholder
| External | Quality | Security risk categories.
For each risk: Risk ID | Category | Description | Trigger Events
| Probability (1-5) | Impact (1-5) | Risk Score (P×I) | Level (Critical/High/Med/Low)
| Owner | Response (Avoid/Transfer/Mitigate/Accept) | Specific Actions | Review Date.
Sort by Risk Score descending. Flag top 3 as priority risks.

PROMPT for RACI Matrix

PROMPT 8 — RACI Matrix
Project: [NAME] | Roles: [PM, Dev Lead, BA, QA, UX, Sponsor, Steering Committee]
Deliverables: [LIST FROM WBS]
Build a complete RACI matrix. Rules to follow:
- Exactly ONE Accountable per row (never two)
- At least one Responsible per row - Consulted used sparingly (max 3 per row)
- No single role dominates all R assignments
Add "Rationale" column for any non-obvious RACI decisions. Flag any role overload.

Choose the Tool that's Right for You and which you Like 😊



WHAT IS BEST AI FOR DOING CODING?

Right now the consensus top pick is OpenAI's GPT-5.6 Sol for raw coding benchmarks, but it's pricey. For most developers actually shipping code day-to-day, here's how it breaks down:

- Claude Sonnet 5 is the sweet spot for in-editor work (Cursor, etc.), best SWE-bench scores at a reasonable price.
- GPT-5.5 / Codex leads for terminal-driven agentic workflows and multi-step tasks.
- Gemini 3.1 Pro wins if you need massive context (1M tokens) or front-end generation.
- Cursor and Claude Code are the most popular coding *agents* (model + harness) developers love right now.

My honest take: if you're writing real code in an IDE, Cursor with Claude Sonnet 5 is the combo most devs are happiest with this month. If you want fully autonomous "assign it a task and walk away" style, OpenAI's Codex in ChatGPT is hard to beat.

What kind of coding are you doing? I can narrow it down further.

From Recording to Minutes in 5 Minutes (Project Execution)



1. Capture & Transcribe

Speech-to-text with speaker labels

Teams Intelligent Recap, Zoom AI Companion, Whisper, Otter.ai, Fireflies.ai



2. Summarise & Structure

Transcript → decisions, actions, risks

Copilot, ChatGPT, Claude, Gemini



3. Distribute & Track

Actions pushed to PM tools with owners & dates

Integrations with Asana, Jira, Planner, Monday.com

SAMPLE PROMPT

“Convert this transcript into formal Minutes of Meeting with: Attendees, Key Discussion Points, Decisions Made, Action Items (owner + deadline), and Risks Raised. Use professional project management language.”

30-45 min → seconds

AI drafts; the PM approves. Verify accuracy, owners, and confidentiality — and check recording policy before client meetings.

Four High-Impact Application Areas



Predictive Scheduling

Completion probability, slippage flags, rapid what-if simulation

AI copilots in MS Project / Primavera; ML simulation add-ons



AI Risk Management

NLP scans of documents & comms, dynamic risk scores, early warnings

AI-enabled risk registers; predictive risk platforms



Intelligent Resourcing

Skill-matched allocation, over-allocation alerts, capacity forecasts

AI-assisted resource & capacity planning platforms



Digital Dashboards

Natural-language queries, auto narrative summaries, anomaly alerts

Power BI with Copilot-style AI; PMO reporting platforms

Traditional vs AI-Enabled Methods

TRADITIONAL

AI-ENABLED

Scheduling

Fixed-date Gantt chart, manual updates

Probabilistic forecasts, continuous re-prediction

Risk

Periodic manual risk register reviews

Continuous data-driven scoring & early warnings

Resourcing

Manual allocation spreadsheets

Skill-matched, workload-aware rebalancing

Reporting

Manually compiled status decks

Real-time dashboards, natural-language queries

Illustrative scenario: a predictive tool flags a civil works package four weeks before the slippage would appear on a traditional Gantt — giving the PM time to re-sequence instead of absorbing the delay.

AI Is Only as Good as Its Foundations



Data Quality

Is there enough clean historical project data for predictions to be meaningful?



Change Management

Will the team trust and act on AI-generated flags — or override them out of habit?



Governance

Who is accountable when an AI recommendation is wrong, and how is that reviewed?



Integration

Does the tool connect to existing PM, ERP, and document systems — or create a new silo?



MODULE

3

Workshop: Identifying AI Opportunities

Map your project • Diagnose pain points • Match AI capability • Prioritise

Five Steps to Your First AI Opportunity



Map

List the key stages of a project you currently work on

Diagnose

Identify one recurring pain point or manual task per stage

Match

Pair each pain point with one of the four AI application areas

Prioritise

Plot opportunities on the Impact–Feasibility matrix

Present

Pitch one quick win to your audience or team in under two minutes

Worksheet columns:

Project Stage → Current Challenge → Potential AI Solution → Impact (H/M/L) → Feasibility (H/M/L)

Prioritise: Impact vs Feasibility

Major Project

High value — needs data, budget, or governance investment. Build the business case.

Quick Win

High value, achievable now. Pursue first as an immediate pilot.

Reconsider

Limited value, hard to execute. Deprioritise or discard.

Fill-In

Easy but limited value. Implement opportunistically.



Pattern to expect

Most first quick wins cluster around scheduling or reporting — these draw on data already being collected. AI adoption can start with one well-scoped pilot, not a full transformation.

FEASIBILITY →

From Half-Day to the Full 3-Day Programme

Day 1

Foundations & Strategy

Deep dive into AI/ML concepts, lifecycle mapping, organisational readiness assessment

→ *A shared vocabulary and readiness baseline*

Day 2

Tools Deep-Dive & Labs

Hands-on practice with predictive scheduling, risk analytics, resourcing, and dashboards

→ *Practical fluency with AI-enabled PM tools*

Day 3

Roadmap & Capstone Project

Build a prioritised, resourced AI adoption roadmap for a real project or portfolio

→ *A presentation-ready roadmap for leadership*

Dates, pricing, and delivery format (online / face-to-face / blended) to be confirmed with Sigma Info Analytics Data Sdn Bhd.

Six Things to Take With You

1

PM has evolved through four eras — most organisations blend all of them today. Many companies have embarked using AI and Automated Tools in Project Management

2

Core AI concepts (ML, predictive analytics, NLP, generative AI) need no technical background. If your company is embarking towards Agentic AI and Autonomous AI Project Manager, then it might get technical where you need to be trained or hire a Specialist.

3

AI adds value across the whole lifecycle — initiation through closure

4

Four quick win application areas: scheduling, risk, resourcing, dashboards

5

Adoption succeeds on data quality, governance, and trust — not just the tool

6

Start with one well-scoped quick win, not a full transformation programme